Graham Harper Edwards

— he, him —

Assistant Professor
Earth & Environmental Geosciences
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APPOINTMENTS

Trinity University 2024 →

Assistant Professor

Earth & Environmental Geosciences, Museum Studies, Environmental Studies

Dartmouth College 2021 – 2024

NSF Astronomy & Astrophysics Postdoctoral Fellow 2021–24

Advisors: Prof. C. Brenhin Keller (Earth Sciences) & Prof. Elisabeth R. Newton (Physics & Astronomy)

Lecturer 2023–24

Department of Physics & Astronomy

EDUCATION

University of California Santa Cruz 2021

PhD in Earth Sciences

Applications of the uranium decay systems in deep time and the Quaternary: chronologic insights within planetary interiors and beneath glaciers [PDF]

Committee: Profs. Terrence Blackburn (Chair), Slawek Tulaczyk, Myriam Telus.

Bowdoin College Brunswick, ME, U.S.A. 2014

Bachelor of Arts, Summa cum Laude

Earth and Oceanographic Science with Honors, minor in Classics, ΦBK Society

PUBLICATIONS

* mentored student author

Refereed

12. B. De Sanctis, N.B. Dragone, C. Wanket, C.P. Bueno de Mesquita, S. Crump, G. Piccione, B. Nirula, H. Bender, R. Corbett-Detig, E.T. Rasbury, **G.H. Edwards**, A. Hawthorn, J.J. Welch, A. Rouillard, B. Shapiro, J. Mikucki, T. Blackburn. Ancient metagenomics reveals subglacial microbiomes driven by oxygen availability. *Nature* [in review]. Preprint: doi:10.64898/2025.12.03.692186.

11. R. Rampalli*, J.W. Johnson, M.K. Ness, **G.H. Edwards**, E.R. Newton, E.J. Griffith, M. Bedell, K. Wang. A galactic perspective on the (unremarkable) relative refractory depletion observed in the Sun. *The Astrophysical Journal* [in review]. arXiv:2509.03577.

10. A. Abuawad, M. Griffiths, **G.H. Edwards**, A. Eftekhari, M. El-Ebweini, H. Al-Najar, A. Butmeh, R. Abu Dayyeh, M. El-Shewy, A. Aker. 2025. The ongoing environmental destruction and degradation

of Gaza: the resulting public health crisis. *American Journal of Public Health*, **115**. doi:10.2105/AJPH.2025.308140. SSRN preprint: doi:10.2139/ssrn.5021472.

9. **G.H. Edwards**, G.G. Piccione, T. Blackburn, S. Tulaczyk. 2025. Uranium-series isotopes as tracers of physical and chemical weathering in glacial sediments from Taylor Valley, Antarctica. *Chemical Geology*, **671**. doi:10.1016/j.chemgeo.2024.122463.

8. **G.H. Edwards**, C.B. Keller, E.R. Newton, C.W. Stewart*. 2024. An early giant planet instability recorded in asteroidal meteorites. *Nature Astronomy*, **8**. doi:10.1038/s41550-024-02340-6. arXiv:2309.10906. **Press:** Astrobites.

7. **G.H. Edwards**. 2024. Giant planets migrated shortly after the Solar System's protoplanetary disk dispersed. *Nature Astronomy Research Briefing*. doi:10.1038/s41550-024-02341-5.

6. R. Rampalli*, M.K. Ness, **G.H. Edwards**, E.R. Newton, M. Bedell. 2024. The Sun remains relatively refractory depleted: elemental abundances for 17,412 Gaia RVS solar analogs and 50 planet hosts. *The Astrophysical Journal*, **965**. doi:10.3847/1538-4357/ad303e. arXiv:2402.16954.

5. M.A. Thompson, M. Telus, **G.H. Edwards**, et al. 2023. Outgassing composition of the Murchison meteorite: implications for volatile depletion of planetesimals and interior-atmosphere connections for terrestrial exoplanets. *Planetary Science Journal*, **4**. doi:10.3847/PSJ/acf760. arXiv:2310.02028.

4. **G.H. Edwards**, T. Blackburn, G. Piccione, S. Tulaczyk, G.H. Miller, C. Sikes*. 2022. Terrestrial evidence for ocean forcing of Heinrich events and subglacial hydrologic connectivity of the Laurentide Ice Sheet. *Science Advances*, **8**. doi:10.1126/sciadv.abp9329.

3. T. Blackburn, **G.H. Edwards**, S. Tulaczyk, M. Scudder*, G. Piccione, B. Hallet, J.C. Zachos, B. Cheney, N. McLean, J.T. Babbe. 2020. Ice retreat in Wilkes Basin of East Antarctica during a warm interglacial. *Nature*, **583**. doi:10.1038/s41586-020-2484-5. **Press:** National Geographic.

2. **G.H. Edwards**, T. Blackburn. 2020. Accretion of a large LL parent planetesimal from a recently formed chondrule population. *Science Advances*, **6**. doi:10.1126/sciadv.aay8641.

1. **G.H. Edwards**, T. Blackburn. 2018. Detecting the extent of ca. 1.1 Ga Midcontinent Rift plume heating using U-Pb thermochronology of the lower crust. *Geology*, **46**. doi:10.1130/G45150.1.

Non-refereed

1. **G.H. Edwards**. 2014. Geochemical and stratigraphic analysis of the Linnévatnet sediment record: a study of Late Holocene cirque glacier activity in Spitsbergen, Svalbard. [*Honor's Thesis*] Bowdoin College: Brunswick, ME, U.S.A. 73 pp. [Undergraduate Research Commons]

TEACHING & MENTORING

Trinity University

Courses Taught

Fall 2024, 2025	GEOS 3410	Global Climate Change
Fall 2025	GEOS 3101	Geoscience Seminar
Spring 2025	GEOS 2402	Earth Surface Processes
Spring 2025	GEOS 1409	Earth's Environmental Systems

Guest Lectures

Fall 2025	ENVI 1301	Introduction to Environmental Studies
Fall 2025	GNED 3151/3252	Ethics, Racism & Gender Inequality in STEM I/II
Fall 2025	INTL 1301	Introduction to International Studies
Fall 2025	GEOS 3407	Planetary Geology

Mentored Undergraduate Researchers

2025 →	Awa Diouf	2025 →	Rose Garrett	2025 →	Luke Moreton
2025 →	Olive Irons	2025 →	Sophie Shipman		

Dartmouth College

Courses Taught

Spring 2024	Exploration of the Solar System
Fall 2023	Exploring the Universe
Fall 2023	The Stretch (Field camp —glacial and climate history section)

Guest Lectures

Winter 2024	Astrobiology
Spring 2022–24	Igneous and Metamorphic Petrology

Research Group Leader 2022 – 2023

Acting leader of a stellar and exoplanetary astronomy research group (4 graduate, 2 undergraduate students) in the Department of Physics & Astronomy while the Primary Investigator was on leave.

Mentored Graduate Researchers

2022 →	Rayna Rampalli (PhD Candidate)	2021–23	Noemi Ortega Dominguez, MSc
2023 →	Jannitta Yao (PhD Candidate)	2021–23	Debbie Sulca, MSc

Mentored Undergraduate Researchers

2023	Jack Duranceau [thesis]	2023	Chase Alvarado-Anderson [thesis]	2022	Cameron Stewart

University of California Santa Cruz

Graduate Teaching Assistant

Spring 2021	Measuring Earth's 4.5 Billion-Year History (isotope geochemistry)
Fall 2020	Evolution of Earth
Fall 2019	Elements of Field Geology
Spring 2018	Geochemistry of the Solar System
Winter 2018	Geologic Principles

Mentored Undergraduate Researchers

2019–21	Cosmo Varah-Sikes [thesis]	2020	Linh Phan	2018–20	Michael Scudder [thesis]
2018	Alexander Levinson	2018	Frances O'Byrne	2017–18	Paul Colosi

Bowdoin College Brunswick, ME, U.S.A.

Laboratory Teaching Assistant

Fall 2012, 2013	Investigating Earth
Fall 2013	Marine Biogeochemistry

Bozeman & Livingston Public Schools MT, U.S.A. 2015

Substitute Teacher

Instructor and paraprofessional in K-12 classrooms.

DEPARTMENTAL SERVICE

Trinity University

Department of Earth & Environmental Geosciences

Committee member: Tenure-track assistant professor of environmental geoscience search 2024
Departmental mentor to first-year faculty, Prof. Xiwei Guo. 2025–2026
Member of *ad hoc* committee developing accessible field experiences for majors. 2025 →

Environmental Studies Program

Contributor to program policy and major requirements.
Guest lecturer in ENVI 1301 (Introduction to Environmental Studies)

Dartmouth College

Department of Earth Sciences

Postdoc Representative for Faculty Search Committee (2 searches) 2023, 2024
Earth & Planetary History Journal Club — Organizer Spring 2022

Department of Earth & Planetary Sciences, UCSC

Whole Earth Seminar — Organizer Spring 2019

INSTITUTIONAL SERVICE

Trinity University

Sustainability Committee 2025 →

Appointed member of TU's standing Sustainability Committee.

Emerging Scholars Commission 2024 →

Developing/implementing career-support and community-building resources for early-career scholars.

Faculty Development Committee 2025

Review panelist for academic leave proposals.

DISCIPLINARY SERVICE

Interdisciplinary Earth Data Alliance 2 (IEDA^2) 2023 →

Community Advisory Board, Chair

- Co-chair of primary community advising body for the NSF-funded IEDA^2 data facility.
- Guide approaches to promote and ensure access, preservation, and re-usability of Earth system data.
- Write annual recommendation reports based on advisory board discussions.

Peer-review referee

Astronomy & Astrophysics, Geochronology, Geology, Nature Communications

Proposal referee

2025 Future Investigators in NASA Earth and Space Science and Technology (FINESST)

Academic Meetings

GSA Connects

2025 Session chair: Asteroid observations, return missions, and meteoritics: interweaving perspectives and data

Lunar & Planetary Science Conference

2025 Dwornik Award Judge

American Astronomical Society

2023 Chambliss Award Judge (best student poster presentation)

Meteoritical Society Annual Meeting

2022 McKay Award Judge (best student oral presentation)

AGU Fall Meeting

2017 Session co-chair: Applications of Thermochronology to Understand Crustal Systems

Society Affiliations

Geochemical Society, Meteoritical Society, National Association of Geoscience Teachers, Geological Society of America, American Geophysical Union, American Astronomical Society.

PUBLIC SERVICE & OUTREACH

Montshire Museum of Science 2022 – 2024

Guest Scientist

- Develop, lead, and assist educational programming and activities within the museum.
- Explain Earth and space science concepts to pre-K through adult audiences.

Santa Cruz Museum of Natural History 2018 – 2022

Volunteer Scientist

- Created educational content and hosted activities at monthly museum events.
- Co-hosted monthly educational video streams covering geoscience topics.

Skype a Scientist 2019–2023

Volunteer Scientist

- Prepare educational content and host video calls with elementary and secondary school classrooms.
- Teach geoscience concepts and share stories about experiences as a scientist.

Geoscientists Encouraging Openness and Diversity in the Earth Sciences (GEODES) 2019 – 2020

Graduate Student Leader

- Student-run group in the UCSC Earth & Planetary Sciences Department.
- Organized and implemented inclusivity-centered outreach and community-building events.

Peary-MacMillan Arctic Museum Brunswick, ME, U.S.A. 2012 – 2016

Student Tour Guide 2012 – 2014

Curatorial Intern 2015 – 2016

- Developed outreach programming, exhibit content, and undergraduate course material.
- Mentored and trained undergraduate student interns and employees.

Museum of the Rockies Bozeman, MT, U.S.A. 2015

Security Volunteer

Answered visitors' questions and facilitated interpretive discussions about natural history exhibits.

LABORATORY & RESEARCH EXPERTISE

Instrumentation

Expert	TIMS
Proficient	MC-ICP-MS, ICP-MS, SEM-EDS
Limited	IRMS, ICP-OES, XRF, XRD

Programming Languages

Expert	Julia
Proficient	MATLAB, Python, R
Limited	HTML, CSS

Laboratory

Geochemistry & Radioisotope Analysis & Computation Laboartory (GRACkLe), Trinity Univ. 2024 →

Primary Investigator — wet chemistry and computational facilities currently in development

W.M. Keck Isotope Facility, UC Santa Cruz 2016 – 2021

Under the advisorship of lab director Prof. Terry Blackburn, I was partially responsible for the operation and maintenance of this multi-user clean lab facility, including:

- Maintenance and operation of mass spectrometery and wet chemistry labs.
- Training and supporting researchers in chemical and mass spectrometric methods.
- Preparation, measurement, and reporting of isotopic analyses of geologic and biomineral material.
- Handling, inventorying, and testing (dosimetry) various radionuclide tracers and standards for mass spectrometric analyses: ^{202}Pb, ^{205}Pb, ^{229}Th, ^{230}Th, ^{232}Th, ^{233}U, ^{235}U, ^{236}U, ^{238}U
- Procurement and preparation of lab consumables.

INVITED ACADEMIC TALKS

11. **The University of Akron** | Invited Research Talk, 6 February 2024
    Title: *Reconstructing and deconstructing ice sheets from the ground up*
10. **Trinity University** | Invited Research Talk, 29 January 2024
    Title: *Reconstructing and deconstructing ice sheets from the ground up*
9. **Colby College** | Invited Research Talk, 1 December 2023
    Title: *Reconstructing and deconstructing ice sheets from the ground up*

8. **Center for Astrophysics | Harvard & Smithsonian** | Astrophysics Data System, 25 October 2023
   Title: *Reconstructing and deconstructing ice sheets from the ground up*

7. **Goldschmidt Conference (Lyon)** | Invited Keynote, 12 July 2023 [abstract]
   Title: *Data & computational science for Earth & planetary history: lessons from the terrestrial archive*

6. **Center for Astrophysics | Harvard & Smithsonian** | Astrophysics Data System, 13 June 2023
   Title: *Transcribing the cosmochemical codex*

5. **Colby College** | Geology Seminar, 17 Mar 2023
   Title: *In the light of a newborn sun: transcribing the first chapters of our solar system from meteorites*

4. **Thayer School of Engineering, Dartmouth College** | Ice+Climate Seminar Series, 10 Feb 2023
   Title: *A terrestrial record of Heinrich events*

3. **Carnegie Institution for Science** | Astronomy Seminar Series, 28 Oct 2022
   Title: *Reconstructing asteroid assembly from the thermal records of chondrites*

2. **Massachusetts Institute of Technology** | Planetary Lunch Seminar, 3 May 2022
   Title: *The early history of the LL chondrite parent planetesimal*

1. **Princeton University** | Environmental Geology and Geochemistry Seminar, 31 Mar 2022
   Title: *Peering beneath the northern Laurentide ice sheet during the last glacial maximum*

INVITED OUTREACH TALKS & COMMUNITY ENGAGEMENT

6. **Dartmouth College** | 8 April, 2024
   Featured scientist for eclipse-viewing in Hanover, NH. (Photo from event)

5. **Montshire Museum of Science** | 7 April, 2024
   Guest scientist presentation on the science of eclipses.

4. **Woodstock Union Middle School** | 29 February, 2024
   Guest scientist presentation and discussion with grade 8 science students.

3. **Moosilauke Ravine Lodge** | 11 August, 2023
   Guest scientist & speaker for Perseid meteor shower. Public talk for families.

2. **Richmond Middle School** | 5 April, 2023 (Hanover, NH)
   Guest scientist presentation for all grade 6 students.

1. **Montshire Museum of Science** | 21 Januray, 2023
   Public talk on meteorites at "Astronomy Day" event (middle to high school learning level).

FUNDING

2025	$9 000	Trinity University	Summer Research Stipend
2021–24	$310 000	National Science Foundation	Award #2102591
2020–21	$10 000	ARCS Foundation	Scholar Award
2016–17		Univ. of California Santa Cruz	Regent's (Graduate) Fellowship
2013	$2 000	Bowdoin College	Grua/O'Connell Research Award
2012	$3 700	Bowdoin College	Freedman Research Fellowship

AWARDS & HONORS

2024	A. Lincoln Washburn Award for Outstanding Mentoring, Dartmouth Earth Sciences
2020	UCSC Earth & Planetary Sciences Outstanding TA Award (Honorable Mention)
2019	Aaron and Elizabeth Waters Award
2014	National Association of Geoscience Teachers Outstanding TA Award
2011–13	Sarah and James Bowdoin Scholar (Dean's List)

IN POPULAR MEDIA

2024	Sky & Telescope	Meteorites tighten timeline for giant planets' movement…
2024	Science Magazine	Giant planets ran amok soon after solar system's birth
2020	National Geographic	Biggest ice sheet on Earth more vulnerable to melting than thought

SELECTED CONFERENCE ABSTRACTS

More comprehensive lists of abstracts are available on the Astrophysical Data System and Google Scholar.

* mentored student author

10. E.R. Garrett*, L.P. Moreton*, **G.H. Edwards**, G. Piccione. 2025. [Poster]: Modern problems require ancient lakes: the chronology and chemistry of Glacial Lake Hitchcock. *Geological Society of America: Connects 2025*, San Antonio, TX, U.S.A. Abstract No. 9078.

9. **G.H. Edwards**, G. Piccione, L.P. Moreton*, E.R. Garrett*, D.S. Jones, C. Danhof. 2025. [Oral]: A hydrologically and biologically complex proglacial lake system at the margins of the collapsing Laurentide Ice Sheet. *Geological Society of America: Connects 2025*, San Antonio, TX, U.S.A. Abstract No. 8850.

8. **G.H. Edwards**, C.B. Keller, E.R. Newton, R. Rampalli*. 2025. [Poster]: Statistical cosmochemistry: gaining new perspectives from old (and new) data. *56th Lunar & Planetary Science Conference*, Woodlands, TX, U.S.A. Abstract No. 2505.

7. **G.H. Edwards**, C.B. Keller, E.R. Newton, C. Stewart*. 2023. [Talk]: Disk-driven migration of Jupiter: support from meteorite thermochronology. *Goldschmidt Conf.*, Lyon, FR. doi:10.7185/gold2023.18450

6. **G.H. Edwards**, G. Piccione, M. Gomez. 2022. [Poster]: Comparing in-person and virtual modes of a 4-year museum-based geoscience outreach program. *Earth Educators' Rendezvous 2022*, Minneapolis, MN, U.S.A.

5. **G.H. Edwards**, T. Blackburn, G. Piccione, S. Tulaczyk, G.H. Miller, C. Sikes*. 2020. [Talk]: Baffin Island subglacial precipitates record subglacial melting beneath the northern Laurentide Ice Sheet concurrent with Heinrich events. *American Geophysical Union Fall Meeting*, New Orleans, LA, U.S.A.

4. C.T. Varah-Sikes*, **G.H. Edwards**, T. Blackburn. 2020. [Poster]: Thermal histories in Type 7 ordinary chondrites: interpreting residence in parent body using petrologic observation and Pb-Pb phosphate thermochronology. *Geological Society of America Annual Scientific Meeting*, Virtual.

3. **G.H. Edwards**, T. Blackburn, S. Tulaczyk, G.G. Piccione. 2019. [Poster]: U-Series isotopics constrain timescale of bedrock comminution and glacial incision in Taylor Valley, Antarctica. *Goldschmidt Conference*, Barcelona, ES.

2. **G.H. Edwards**, T. Blackburn, C.M.O'D. Alexander. 2017. [Talk]: Accretion and disruption histories of the ordinary chondrite parent bodies. *80th Annual Meeting of the Meteoritical Society*, Santa Fe, NM, U.S.A.

1. **G.H. Edwards**. 2014. [Poster]: Geochemical and stratigraphic analysis of the Linnévatnet sediment record: a provenance study of Late Holocene cirque glacier activity in Linnédalen, Spitsbergen, Svalbard. *44th International Arctic Workshop*, Boulder, CO, U.S.A.

Additional Work Experience

Wildland Firefighter (Type 2) — U.S. Forest Service, Winthrop, WA, U.S.A. — 2015.
Server & Bartender — Trio Restaurant, Fish Creek, WI, U.S.A. — 2014.

Non-Academic Interests

Bicycling, cross-country skiing, birdwatching, breadmaking, classical history and literature.